

Assignment 2: Packet Tracer WLAN configuration

Introduction

This assignment focused on the configuration and security setup of wireless networks using Cisco Packet Tracer. The main objective was to simulate a real-world wireless LAN deployment at both a home and enterprise level. In the first part, I configured a wireless router for a home network, adjusting DHCP settings, setting up SSID and wireless security, and connecting various client devices. The second part required setting up a Wireless LAN Controller (WLC) environment, including the creation of two WLANs—one secured with WPA2-PSK and another with WPA2-Enterprise. It also involved configuring VLAN interfaces, DHCP scopes, SNMP settings, and authentication via a RADIUS server. This practical task helped strengthen my understanding of wireless network infrastructure and its security considerations.

PART 1: CONFIGURE A WIRELESS ROUTER

In this scenario, I was installing a new home wireless Router at a friend's house. I needed to change settings on the router to enhance security and meet the friend's requirements.

STEP1: CHANGE DHCP SETTINGS

- a. I opened the Home wireless router GUI and changed the router IP and DHCP settings according to the information in the addressing table:

Device	Interface	IP Address
Home Wireless Router	Internet	DHCP
	LAN	192.168.6.1/27
RTR-1	G0/0/0.2	192.168.2.1/24
	G0/0/0.5	192.168.5.1/24
	G0/0/0.100	192.168.100.1/24
	G0/0/1	10.6.0.1/24
SW1	VLAN 200	192.168.100.100/24
LAP-1	G0	DHCP
WLC-1	Management	192.168.100.254/24
RADIUS Server	NIC	10.6.0.254/24
Home Admin	NIC	DHCP
Enterprise Admin	NIC	192.168.100.200/24
Web Server	NIC	203.0.113.78/24
DNS Server	NIC	10.100.100.252
Laptop	NIC	DHCP
Tablet PC	Wireless0	DHCP
Smartphone	Wireless0	DHCP
Wireless Host 1	Wireless0	DHCP
Wireless Host 2	Wireless0	DHCP

- b. I then permitted a maximum of 20 addresses to be issued by the router.
- c. Next step was to configure the DHCP server to start with IP address 3 of the LAN network.
- d. Next I configured the internet interface of the router to receive its IP address over DHCP. I verified the address and it was 192.168.0.1 , I changed it to be according to the Addressing table 192.168.6.1
- e. Next step was to configure the static DNS server to the address in the Addressing table

After setting that up this is how it looks:

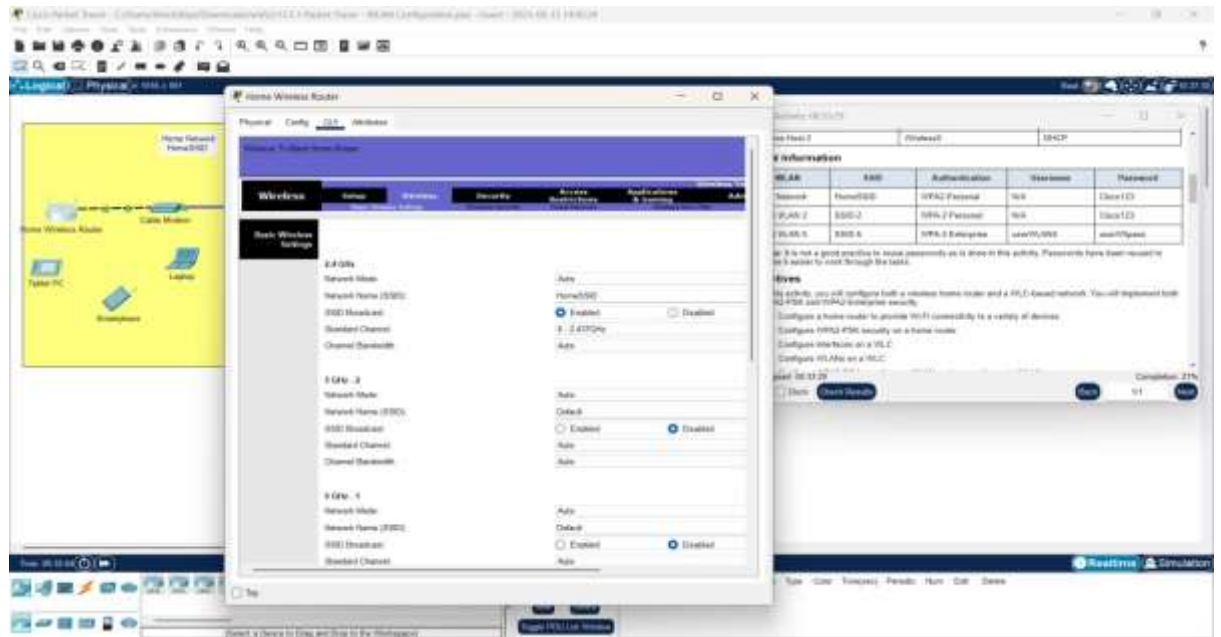


STEP 2: Configuring the Wireless LAN

- a. The Network will use the 2.4GHz wireless LAN interface. I therefore configured the interface with the SSID shown in the wireless LAN information table below:

WLAN	SSID	Authentication	Username	Password
Home Network	HomeSSID	WPA2-Personal	N/A	Cisco123
WLAN VLAN 2	SSID-2	WPA-2 Personal	N/A	Cisco123
WLAN VLAN 5	SSID-5	WPA-2 Enterprise	userWLAN5	userW5pass

- b. used channel 6 as the standard channel
- c. Being sure that all wireless hosts in the home will be able to see the SSID. And the configuration for this part appeared this way:

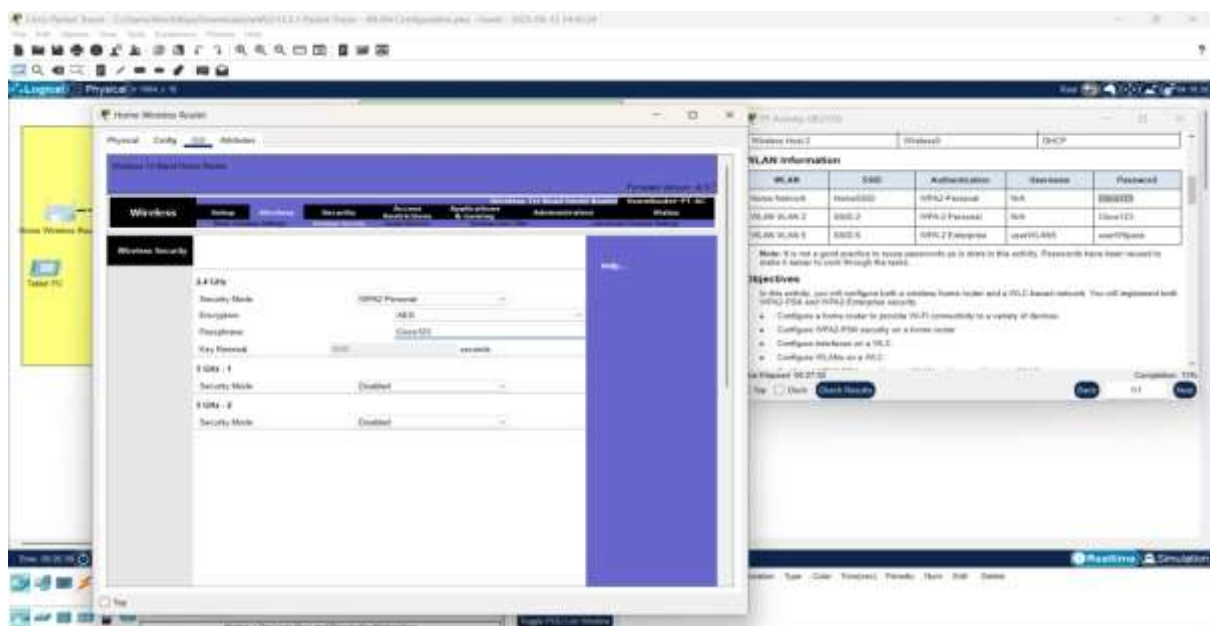


STEP 3: Configure security

This step had 2 major tasks

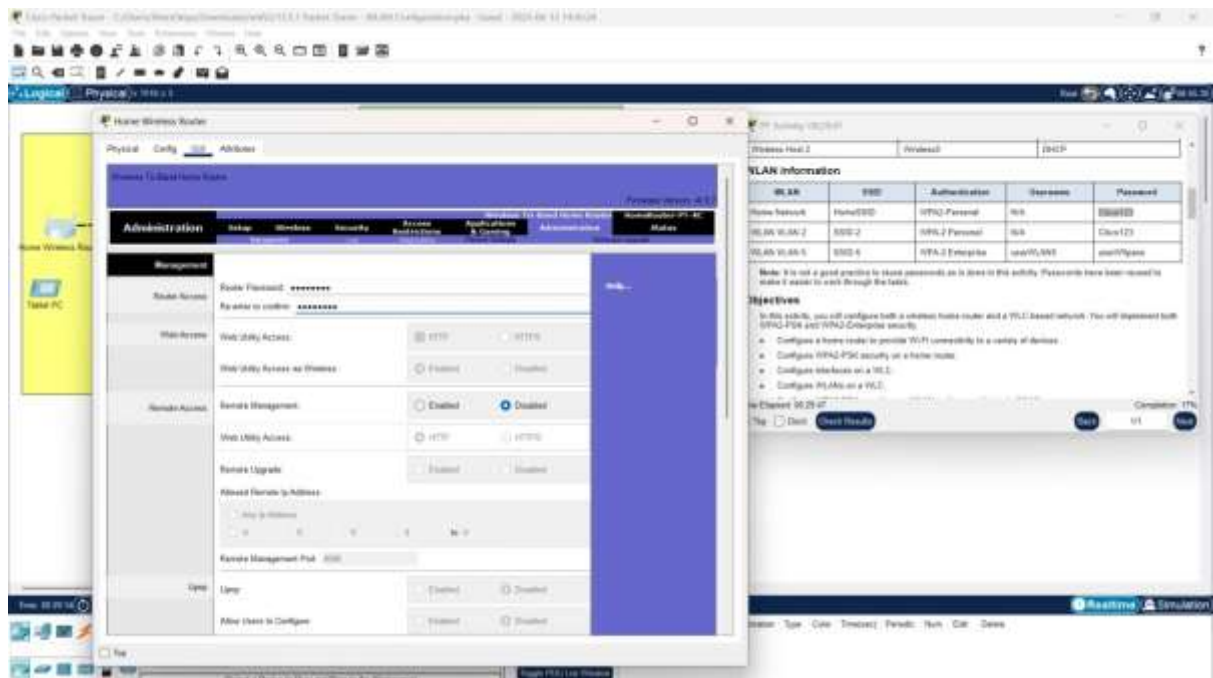
1. Configuring wireless LAN security. I used WPA2 Personal and the passphrase shown in the wireless LAN information table.

To do this I went to the wireless security settings under the wireless section and configured these settings on 2.4GHz settings as shown below



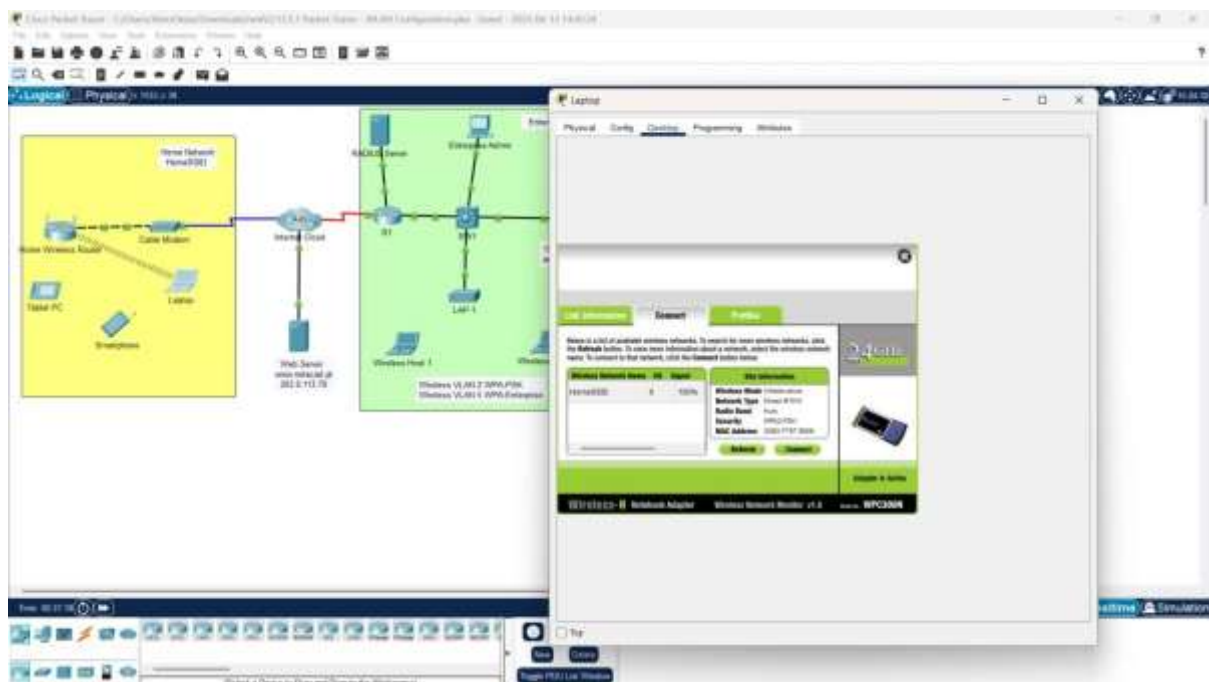
2. I then secured the router by changing the default password to the value shown in the Wireless LAN information table

To do this I went to the Administration section of the router and removed the defaults on Router password section and entered the given credentials

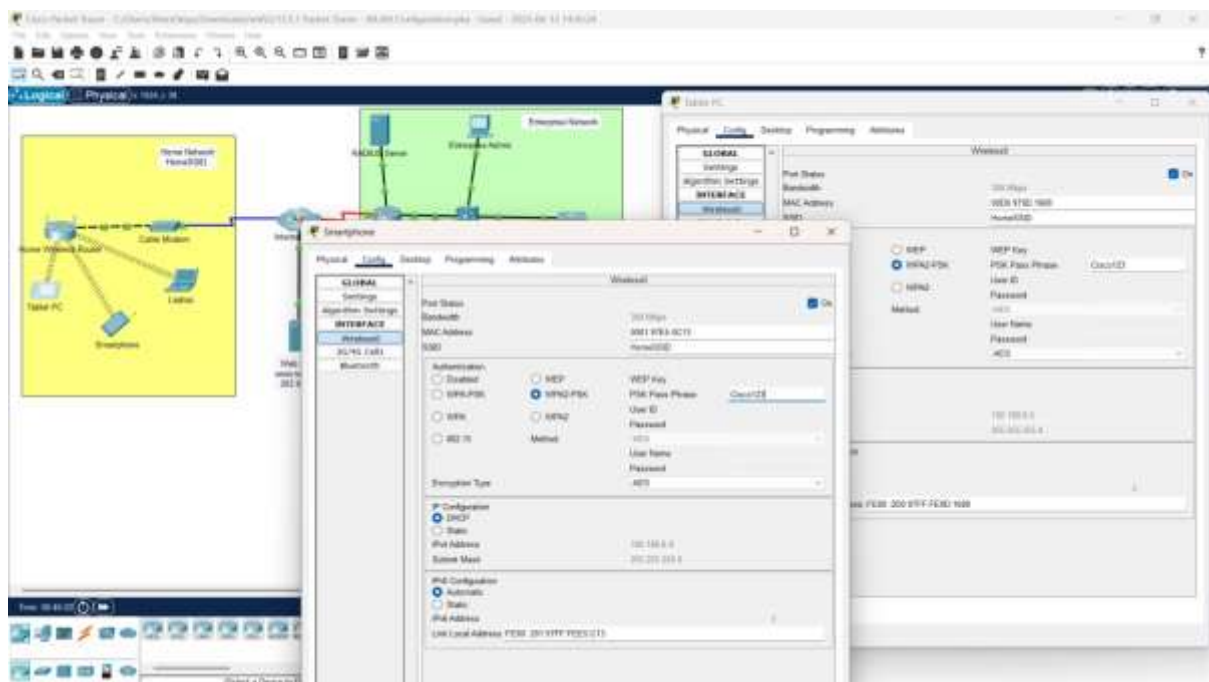


STEP 4: Connect Clients to the Network

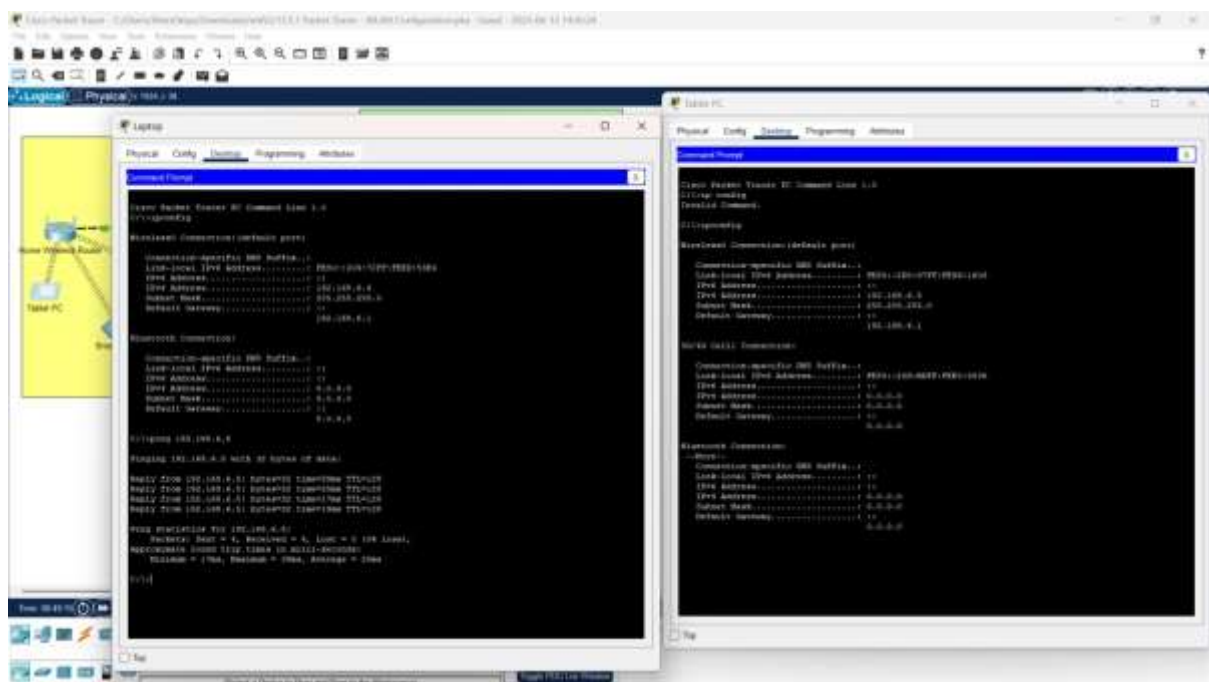
1 I opened the pc wireless app on the desktop of the laptop and configured the client to connect to the network.



2 I opened the config tab on the tablet pc and smartphone and configured the wireless interfaces to connect to the wireless network.



3 I then verified the connectivity. The hosts were able to ping each other.



Pinging the web server was also successful as shown below:

2. I then proceeded to configure an interface for the first WLAN With the following details:

Name: WLAN 2

VLAN Identifier: 2

Port Number: 1

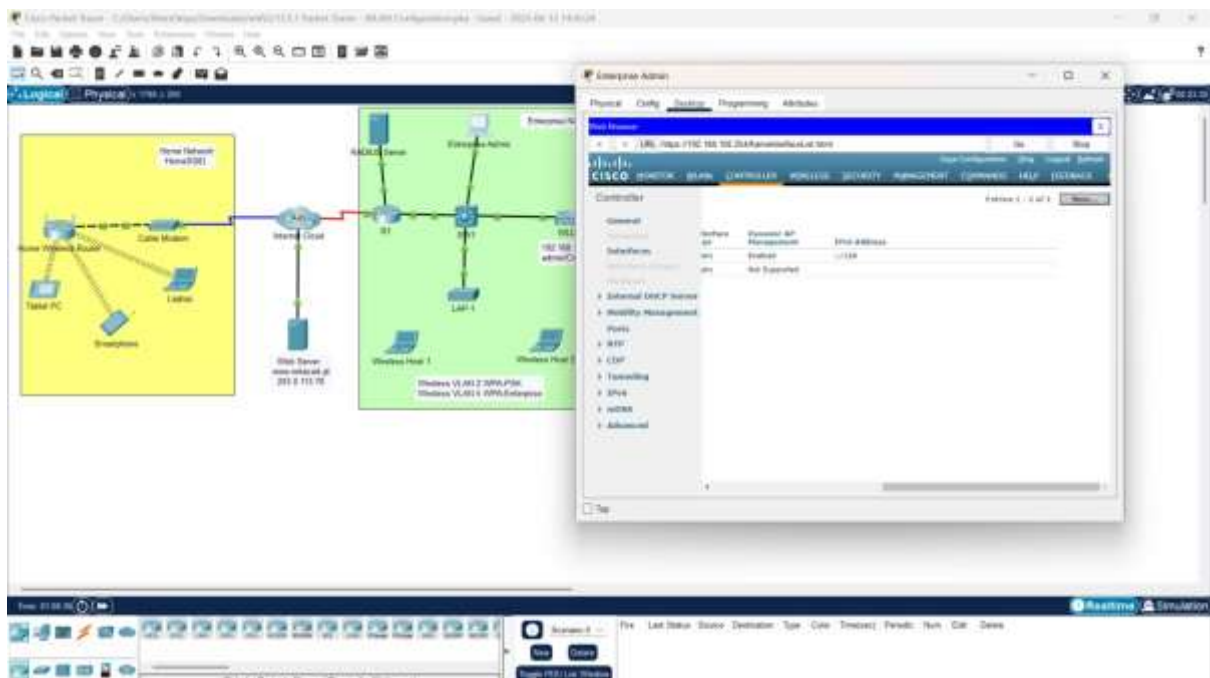
Interface IP Address: 192.168.2.254

Netmask: 255.255.255.0

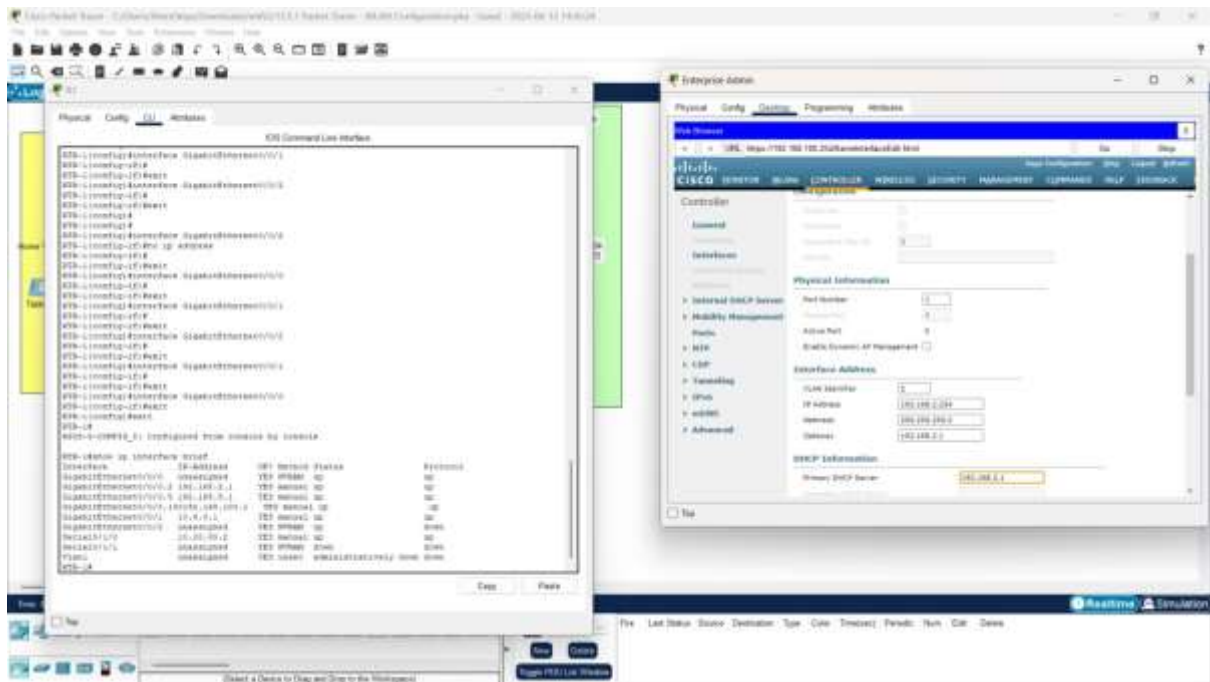
Gateway: RTR-1 G0/0/0.2 address

Primary DHCP Server: Gateway address

To start I clicked on the Controller tab, then Interfaces, then New



I then used the show interface brief to know the ip of G0/0/0.2 interface which turned out to be 192.168.2.1 then configured I the first interface as described:



3. I then configured an interface for the second WLAN:

Name: WLAN 5

VLAN Identifier: 5

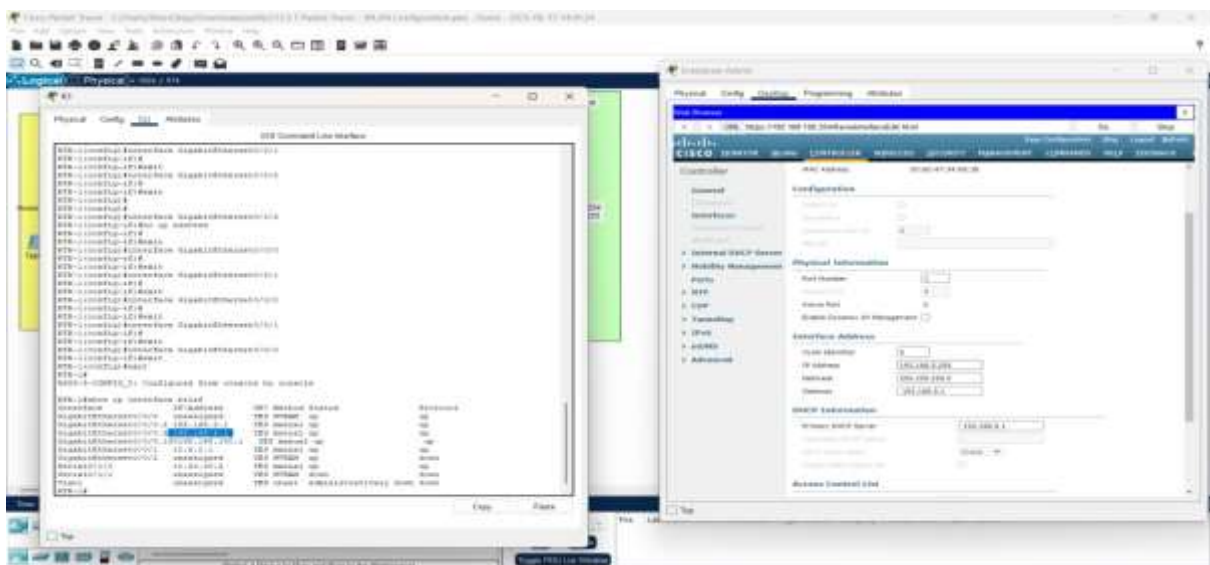
Port Number: 1

Interface IP Address: 192.168.2.254

Netmask: 255.255.255.0

Gateway: RTR-1 interface G0/0/0.5 address

Primary DHCP Server: Gateway address which I found out to be 192.168.5.1



And the output was this:

The screenshot shows the Cisco Enterprise Admin web interface. The top navigation bar includes tabs for Physical, Config, Desktop, Programming, and Attributes. Below this is a Web Browser window showing the URL <https://192.168.100.254/frameInterfaceList.html>. The main content area is titled 'Controller' and 'Interfaces'. The left sidebar lists various configuration options under the 'Controller' tab, with 'Internal DHCP Server' selected. The main area displays a table of interfaces.

Interface Name	VLAN Identifier	IP Address	Interface Type	Dynamic AP Management
WLAN 2	2	192.168.2.254	Dynamic	Disabled
WLAN 5	5	192.168.5.254	Dynamic	Disabled
management	1	192.168.100.254	Static	Enabled
virtual	N/A	192.0.2.1	Static	Not Supported

STEP2: Configure a DHCP scope for the wireless management network

Next was to Configure and enable an internal DHCP Scope as follows:

Scope Name : Management

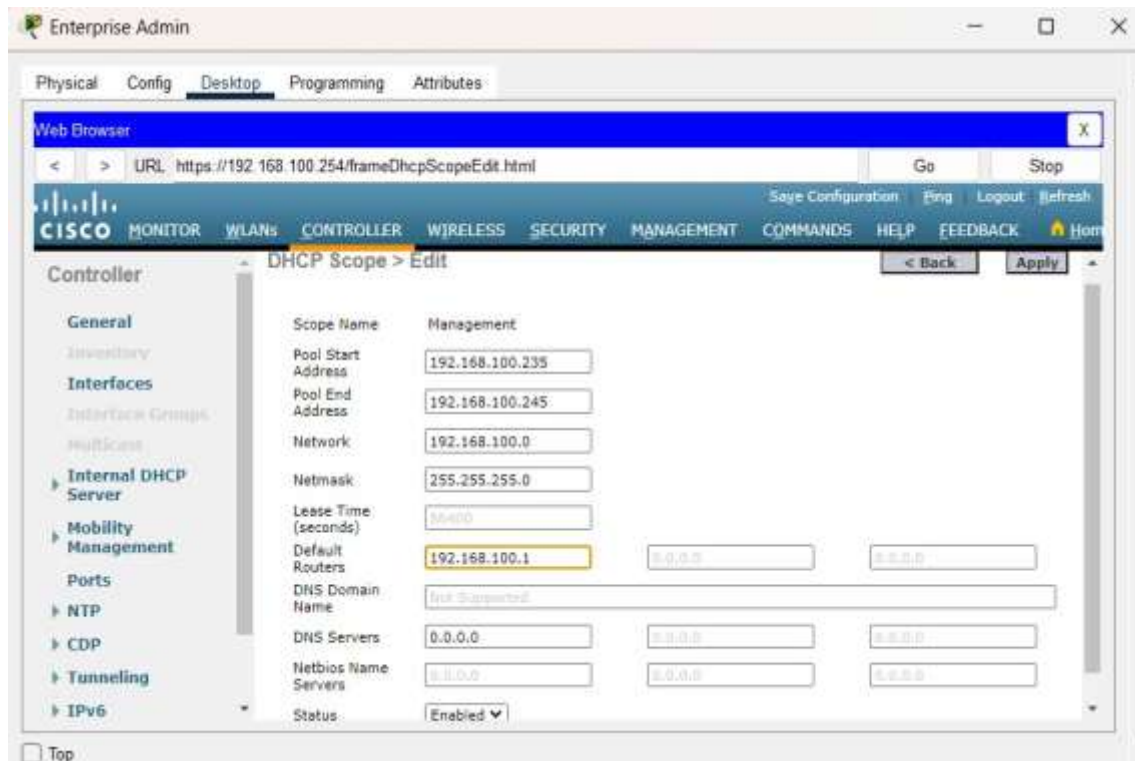
Pool Start Address: 192.168.100.235

Pool End Address: 192.168.100.245

Network: 192.168.100.0

Netmask: 255.255.255.0

Default Routers: 192.168.100.1



STEP3: Configure the WLC with external server address

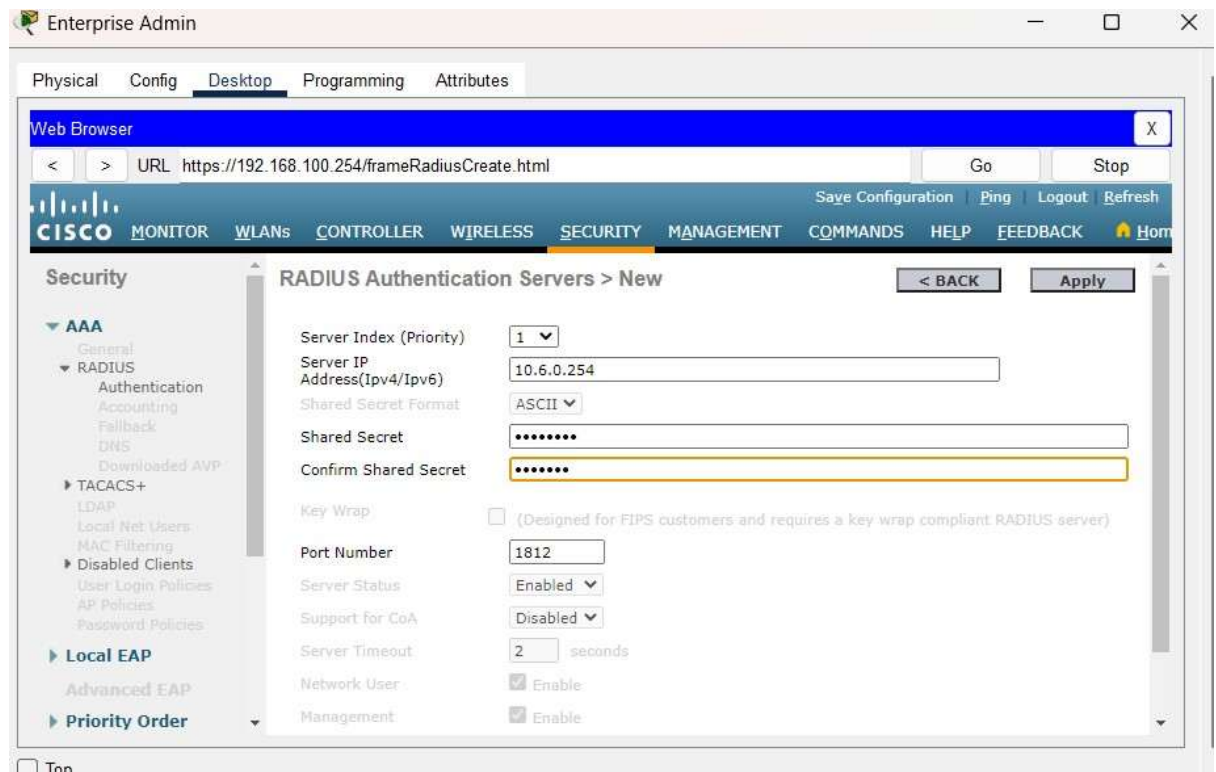
To do this I navigated to the Security tab and clicked on new to configure a new Radius server

- a. Next I configured the radius server information as follows:

Server Index: 1

Server Address: 10.6.0.254

Shared Secret: Radius PW

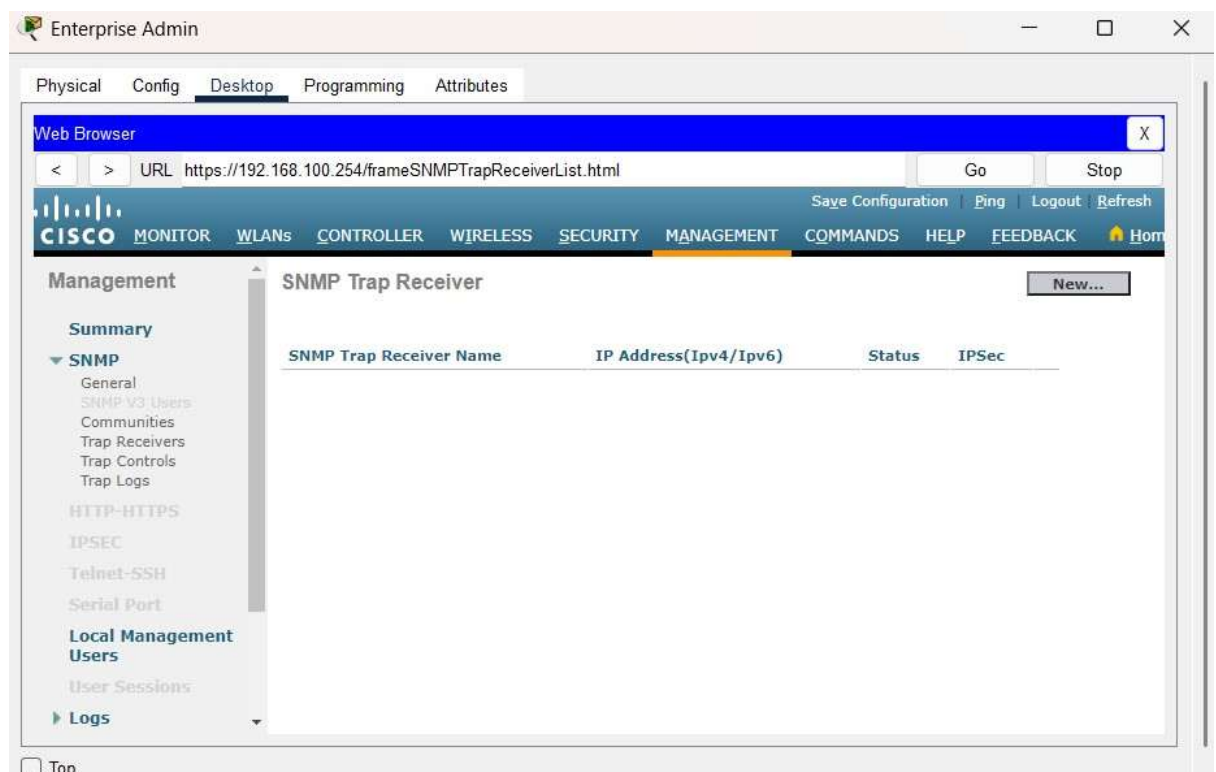


b. Then I configured the WLC to send logs information to an SNMP server:

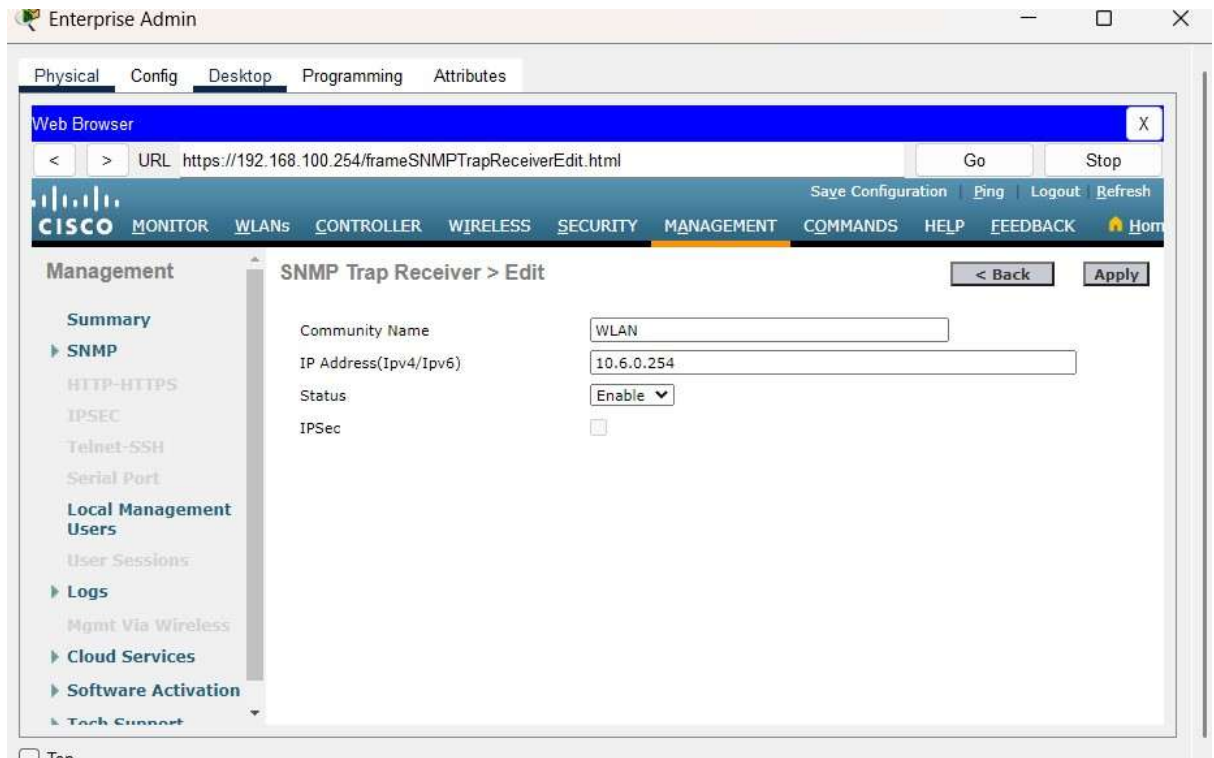
Community Name: WLAN

IP Address: 10.6.0.254

To do this I first went to the management tab then selected SNMP, then Trap receivers then I clicked New



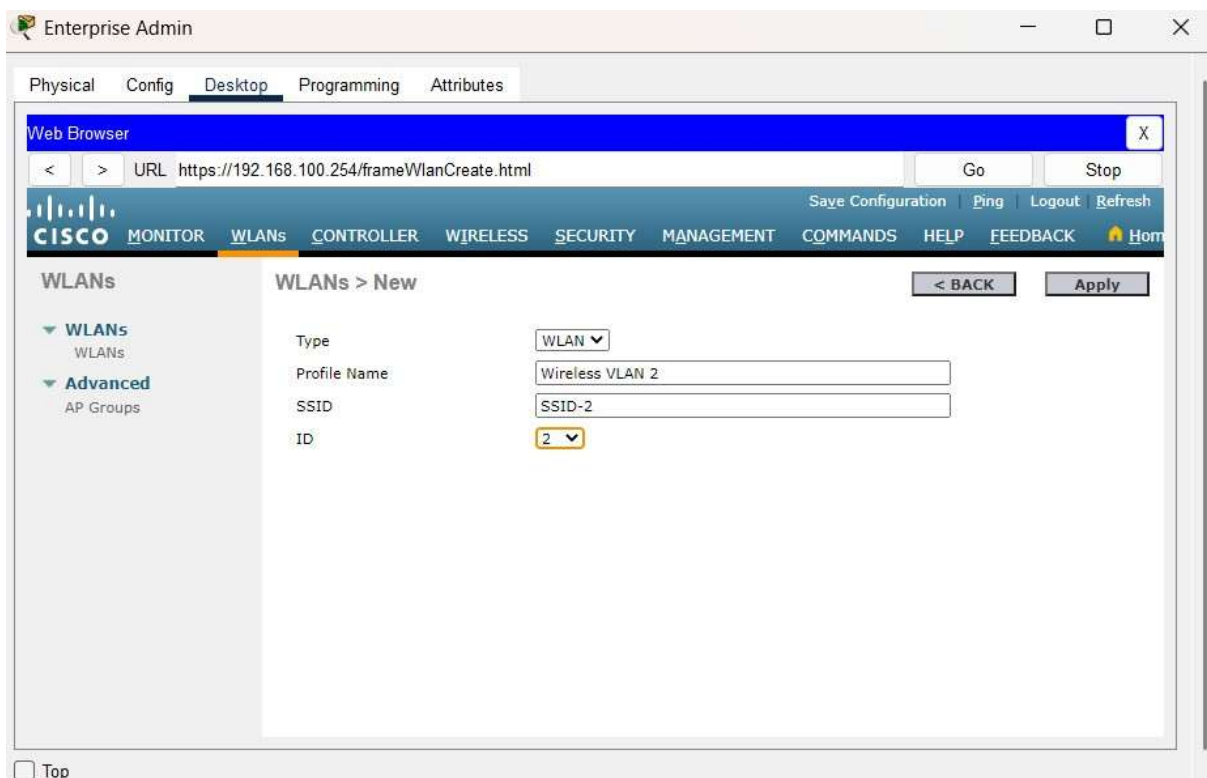
I then added the configurations:



STEP 4: Create the WLANS

1ST LAN:

To do this I navigated to the WLANs part then clicked on Create New then Go, then i started the configurations:



Profile Name: **Wireless VLAN2**

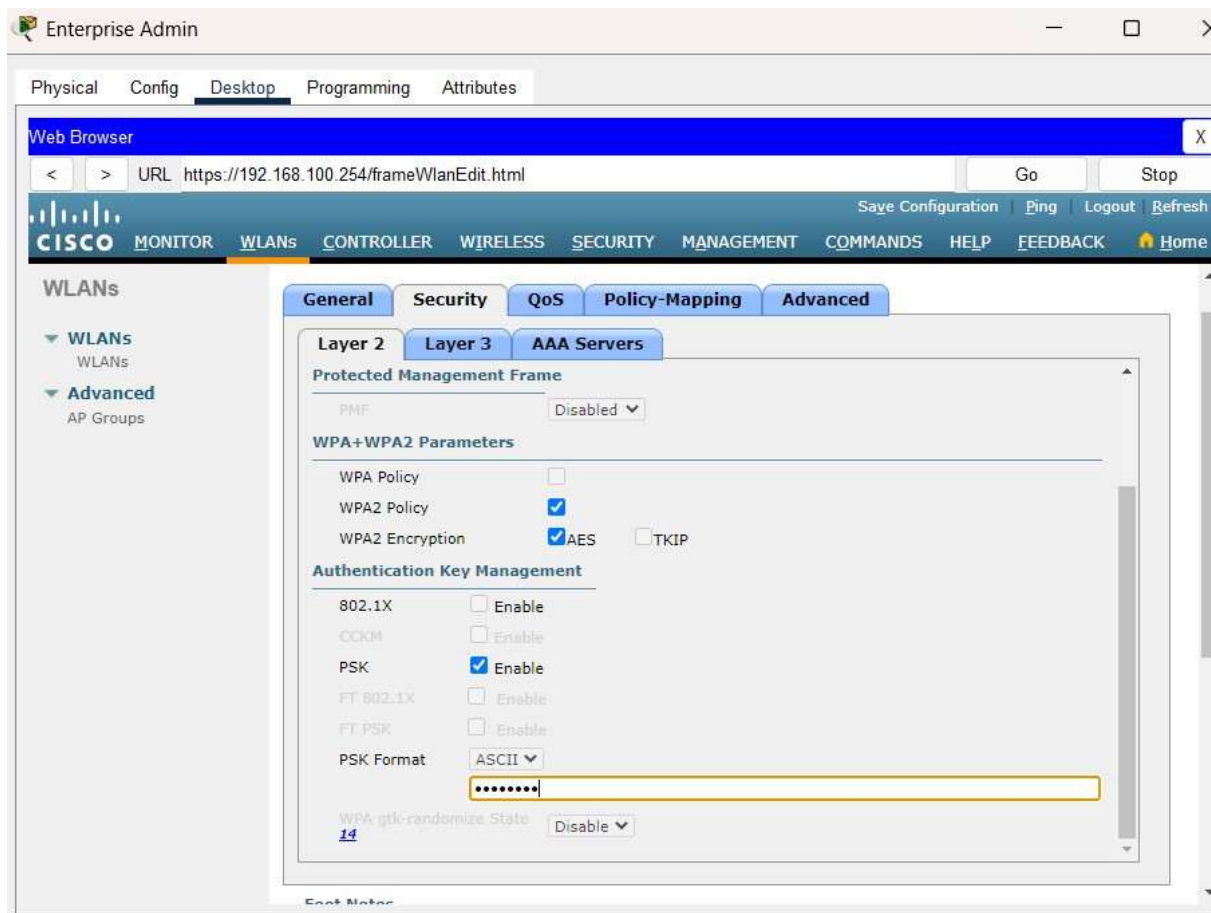
WLAN SSID: **SSID-2**

ID: **2**

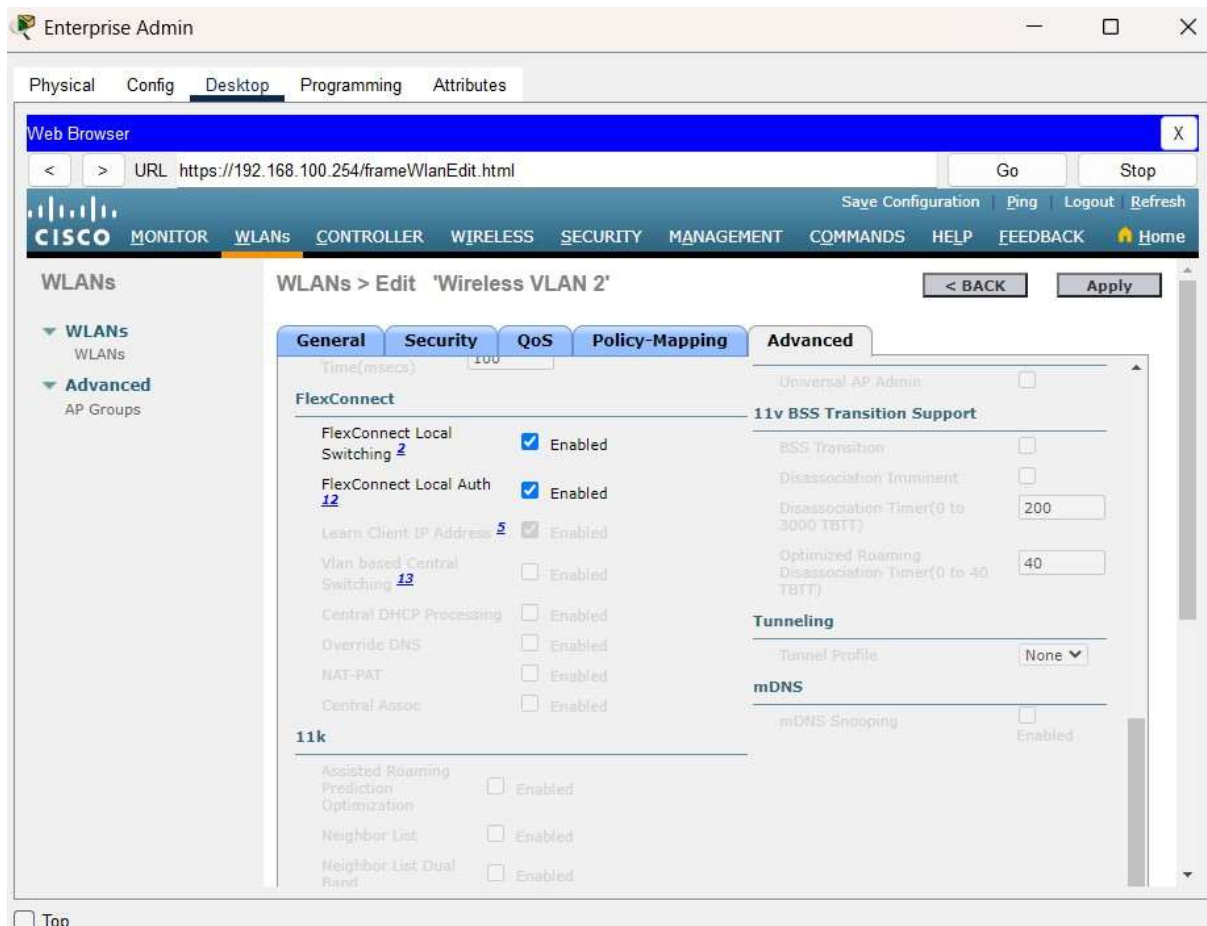
Interface: **WLAN 2**

Security: **WPA2-PSK**

Passphrase: **Cisco123**



Under the advanced tab, I went to the FlexConnect Section. Enabled **FlexConnect Local Switching** and **FlexConnect Local Auth**



2ND LAN:

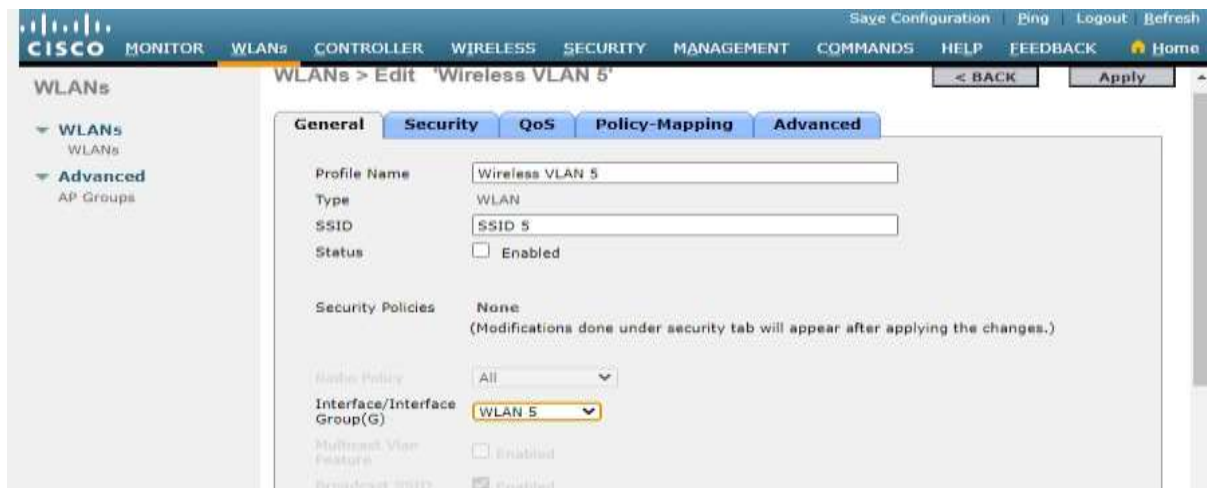
Profile Name: **Wireless VLAN5**

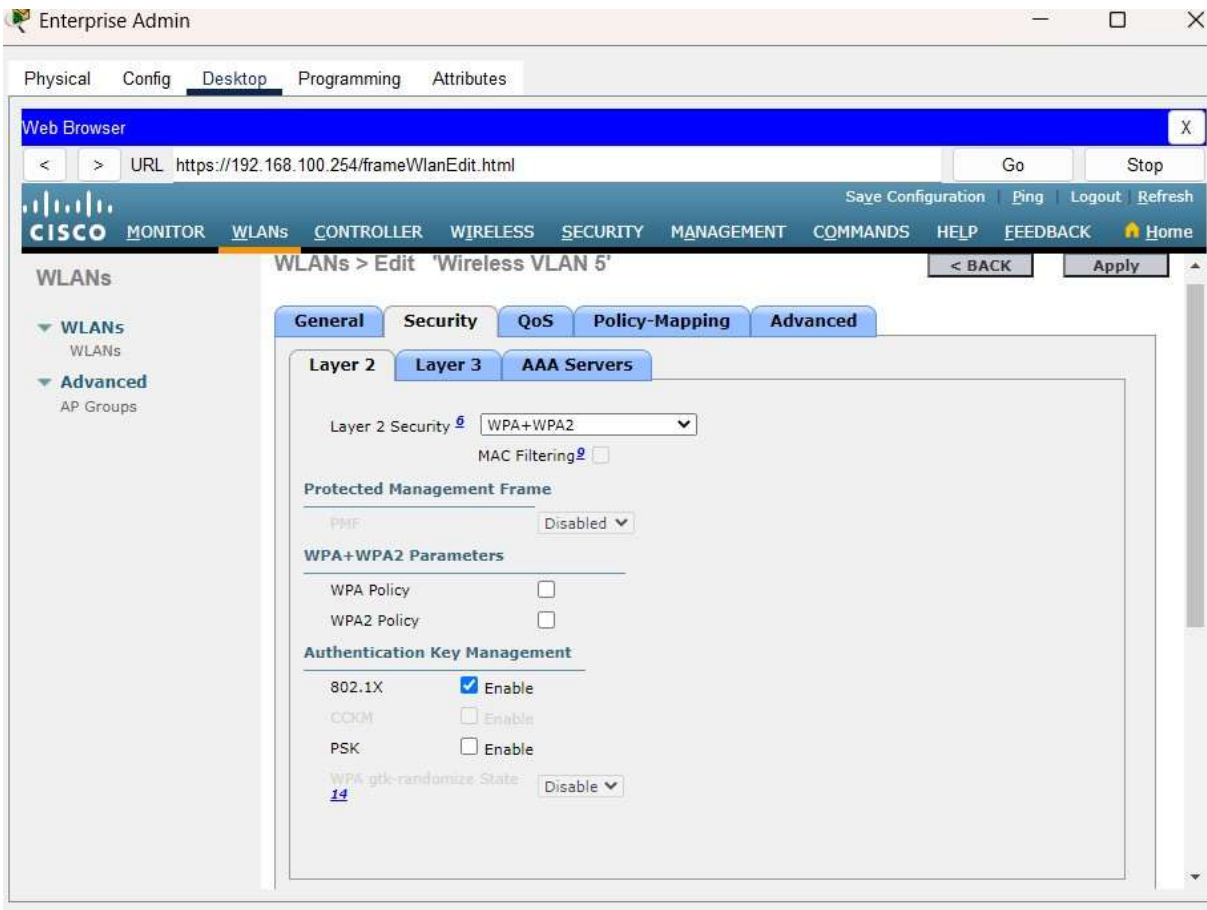
WLAN SSID: **SSID-5**

ID: **5**

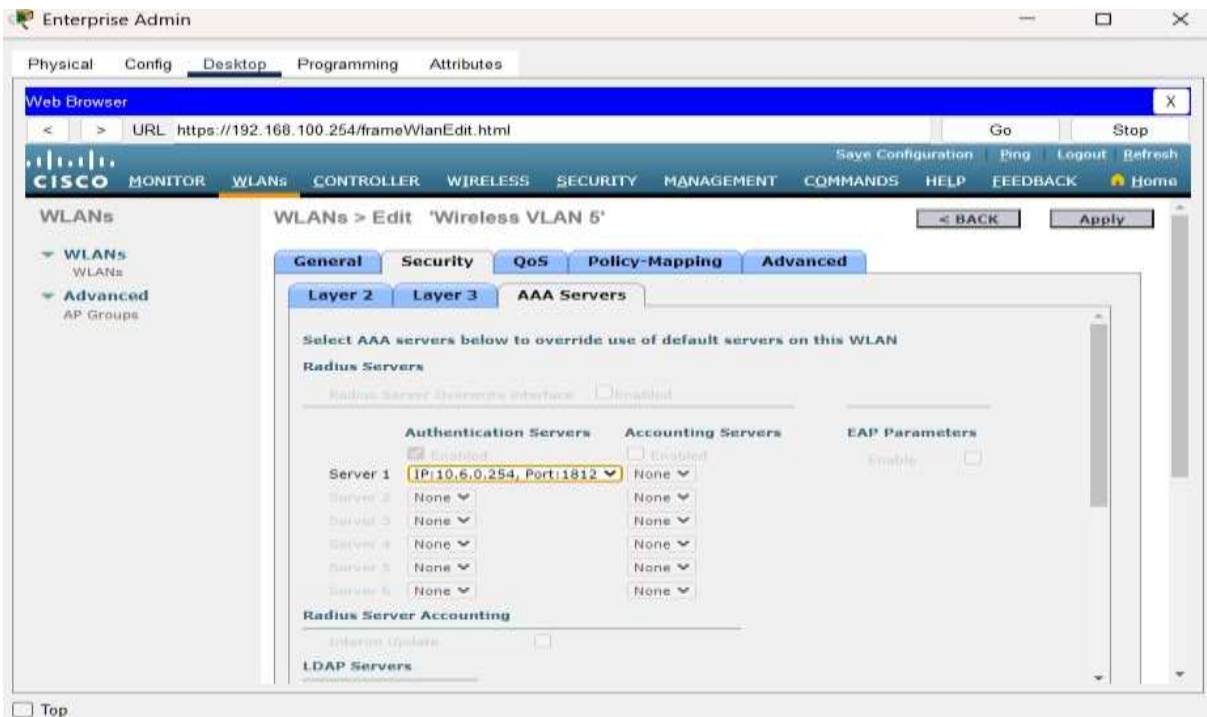
Interface: **WLAN 5**

Security: **802.1x-WPA2-Enterprise**

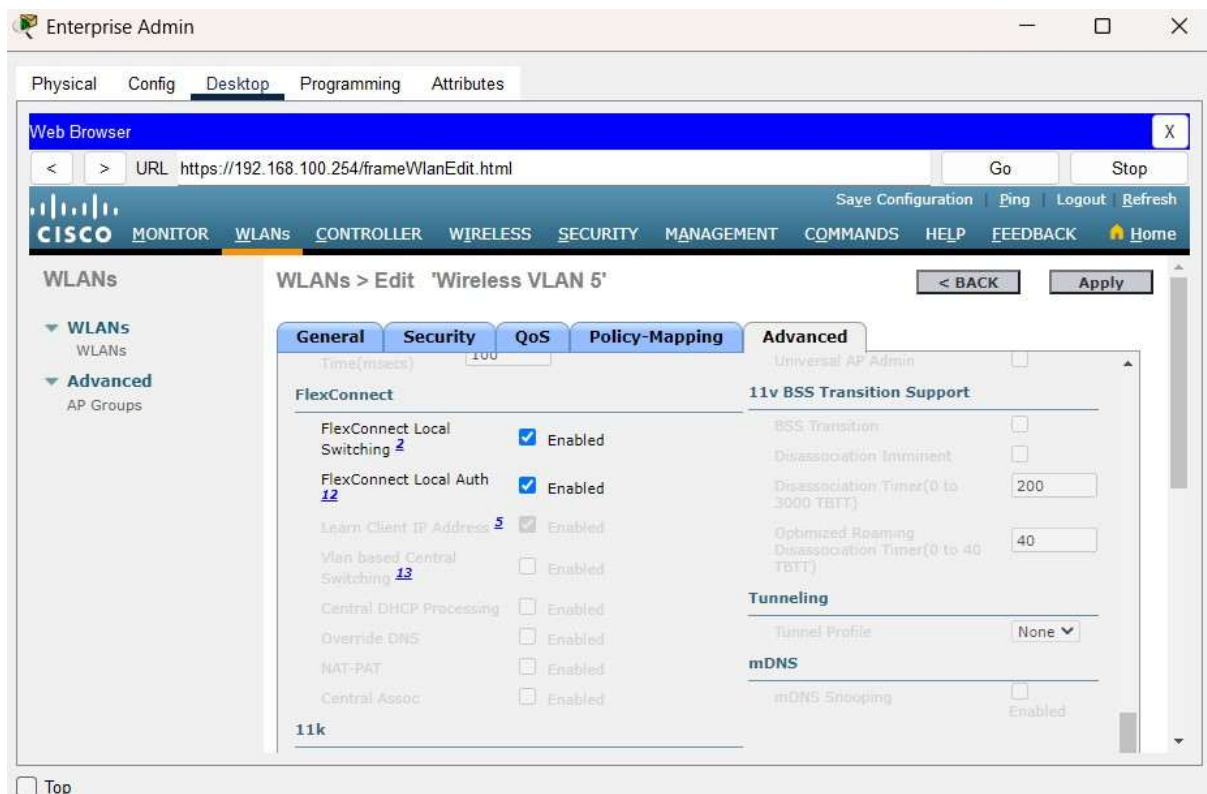




I then configured the WLAN TO USE THE RADIUS server for authentication.



Then I made the **FlexConnect** settings as I did on 1st LAN



STEP 5: Configure the Hosts to connect to WLANs.

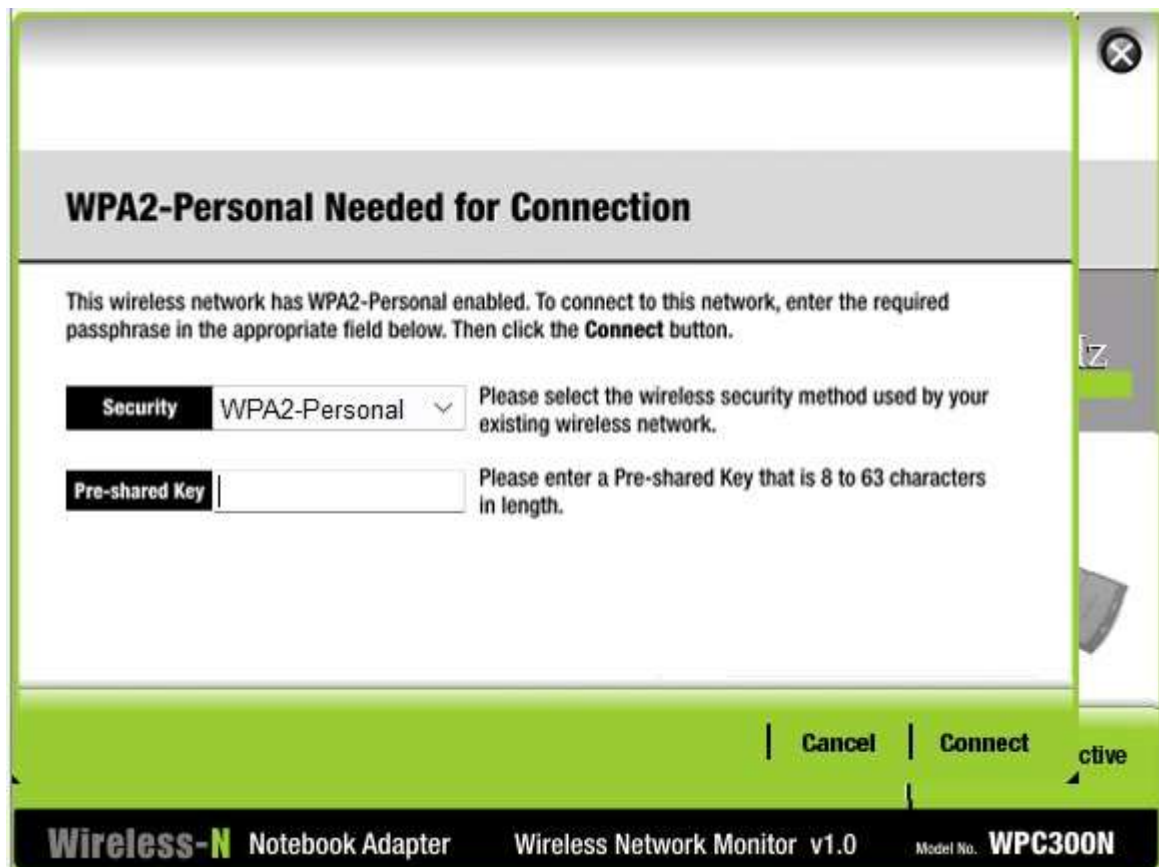
I used the desktop PC wireless app to configure the hosts as follows:

- Wireless Host 1 to connect to wireless VLAN 2.
- Wireless Host 2 to connect to wireless VLAN 5 using the credentials in the WLAN information table.

On the first attempt the SSIDs were not showing, I spent time troubleshooting and found that I had not enabled the SSIDs in WLC2 using the Enterprise Admin pc . Ifirst enabled them in both VLAN 2 and 5

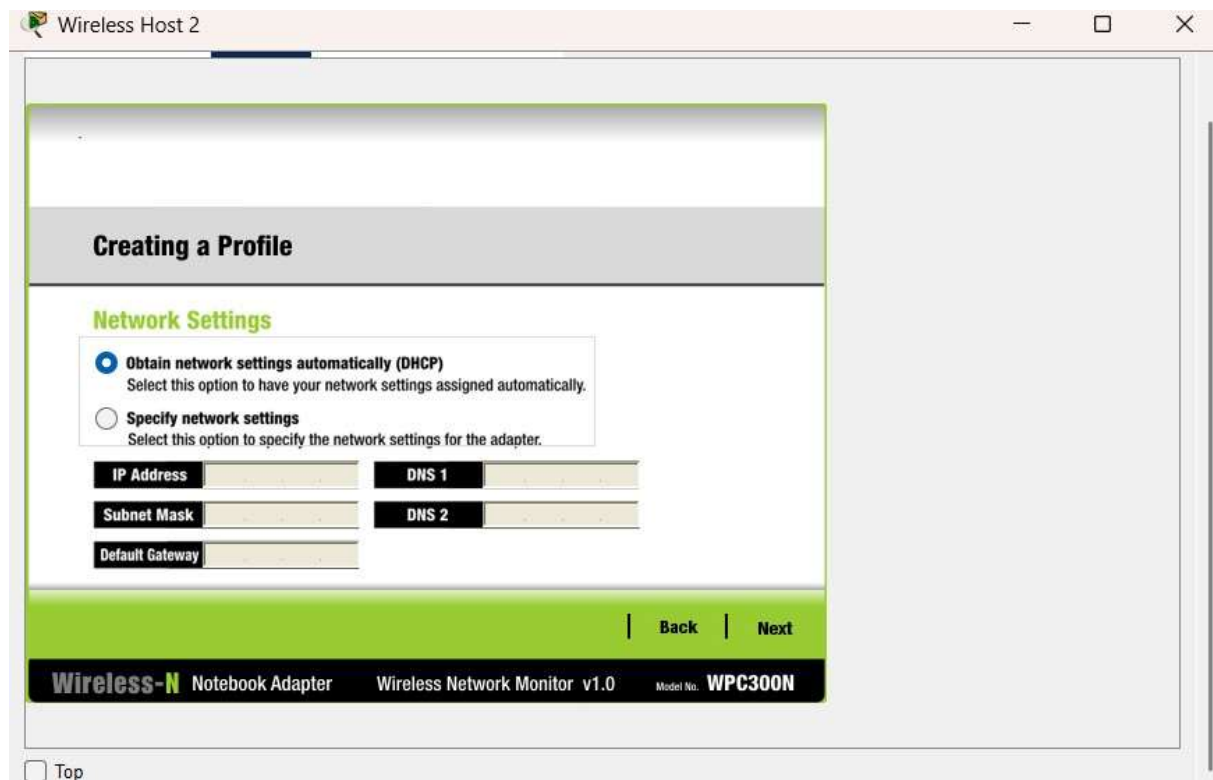
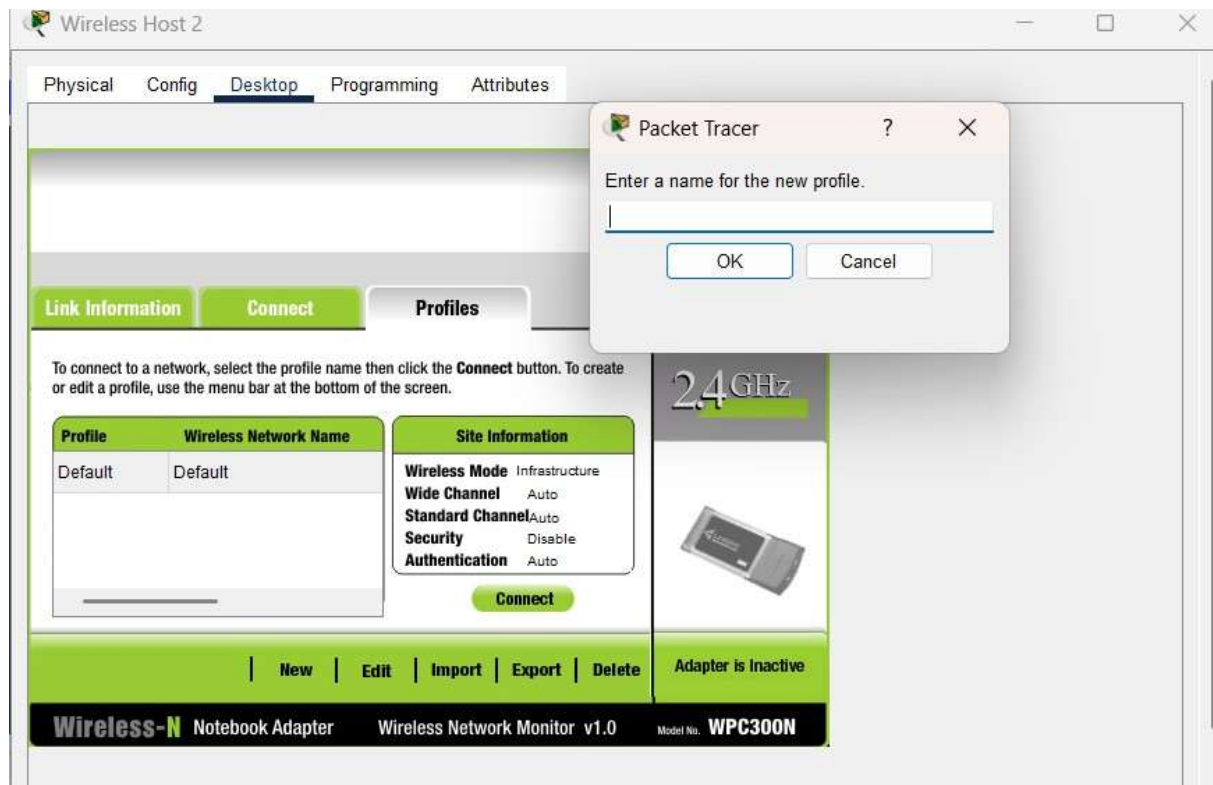


after doing that I was now able to see them and connect to the VLANs



For Wireless host 2, I first had to create a profile before attempting to connect. Below are the screenshots showing each window on each step:





I used the credentials on WLAN Information in this next step:

Wireless Host 2

PhysicalConfigDesktopProgrammingAttributes

Creating a Profile

Wireless Security - WPA2 Enterprise

Authentication	PEAP	Please select the authentication method that you use to access your network.
Login Name	userWLAN5	Enter the Login Name used for authentication.
Password	••••••••	Enter the Password used for authentication.
Server Name		Enter the Server Name used for authentication. (Optional)
Certificate	Trust Any	Please select the certificate used for authentication.
Inner Authen.	TOKEN CARD	Please select the inner authentication method used inside the PEAP tunnel.

BackNext

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Wireless Host 2

PhysicalConfigDesktopProgrammingAttributes

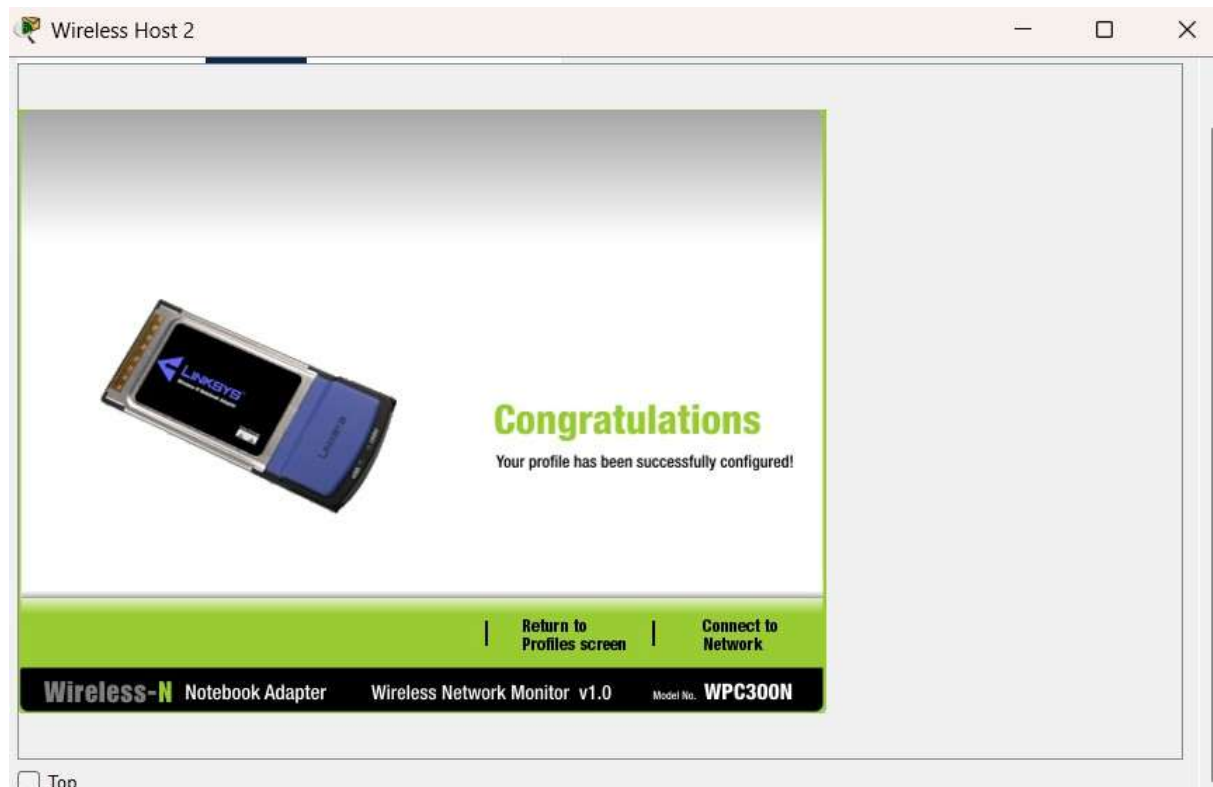
Confirm New Settings

Profile Settings			
Wireless Network Name	SSID 5	IP Address	Auto
Wireless Mode	Infrastructure	Subnet Mask	Auto
Network Mode	Mixed Mode	Default Gateway	Auto
Radio Band	Auto	DNS1	Auto
Wide Channel	Auto	DNS2	
Standard Channel	Auto		
Security	WPA2 Enterprise		
Authentication	Auto		

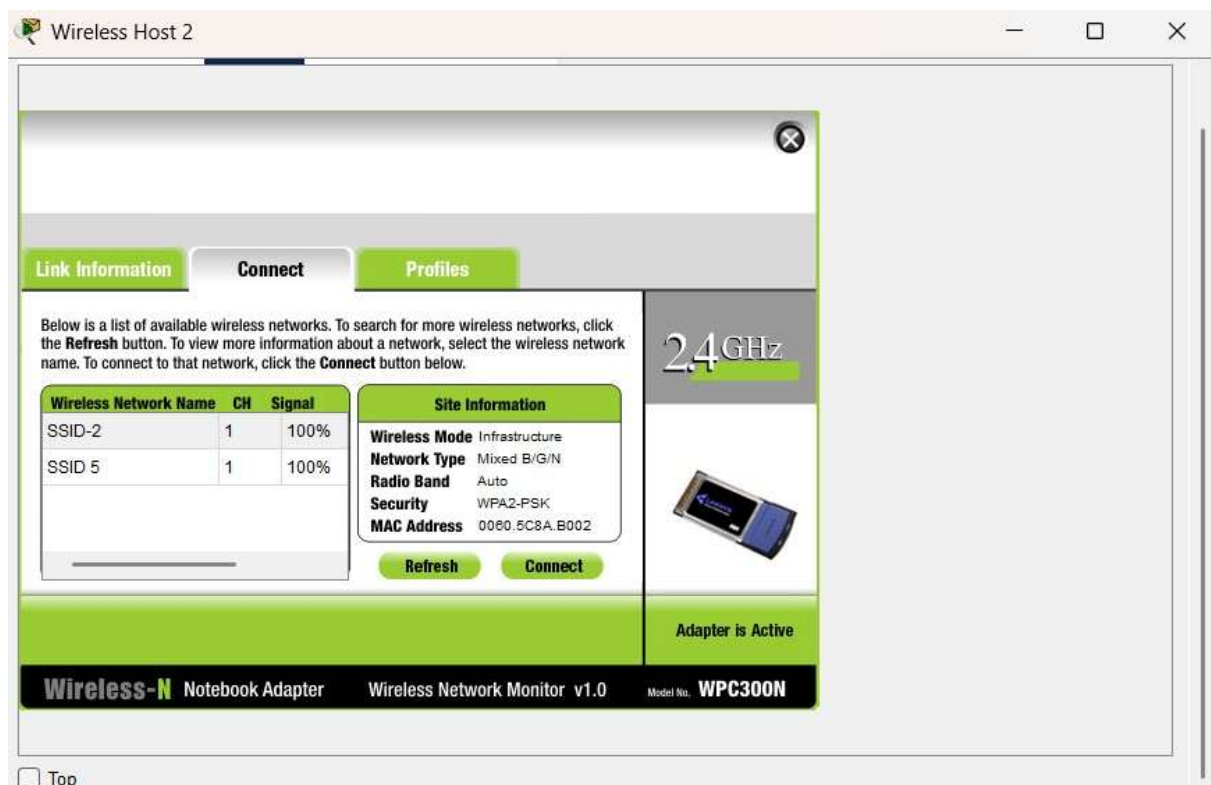
ExitBackSave

Wireless-N Notebook AdapterWireless Network Monitor v1.0Model No. WPC300N

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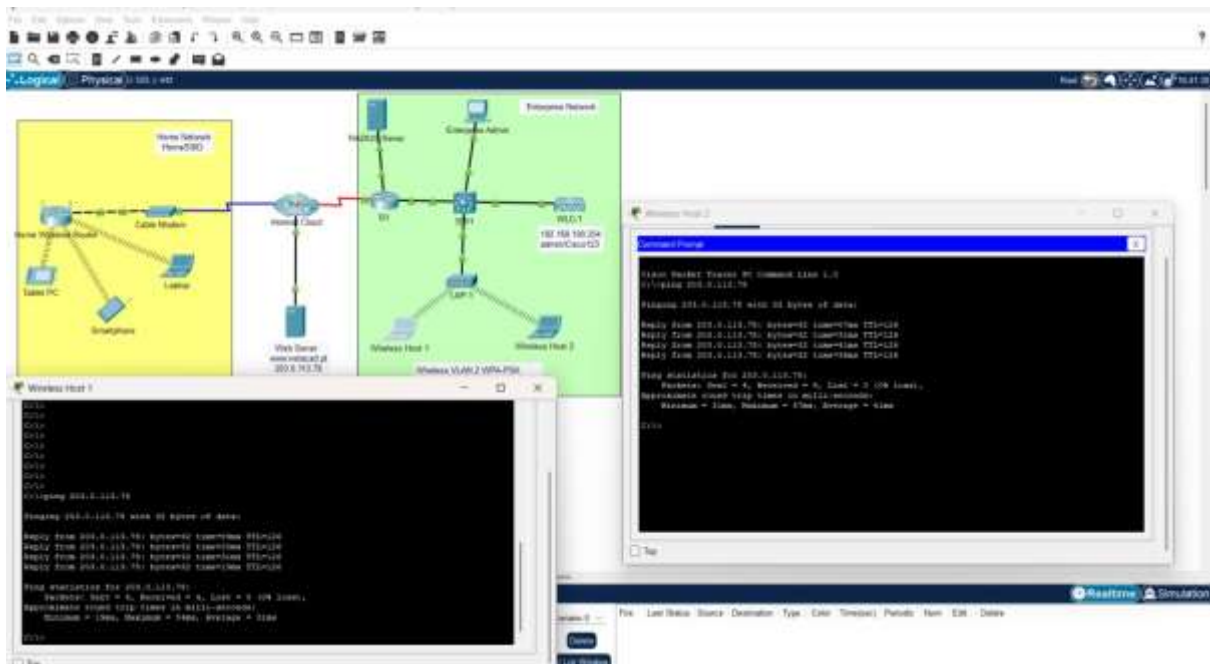


The connection now looked like this:

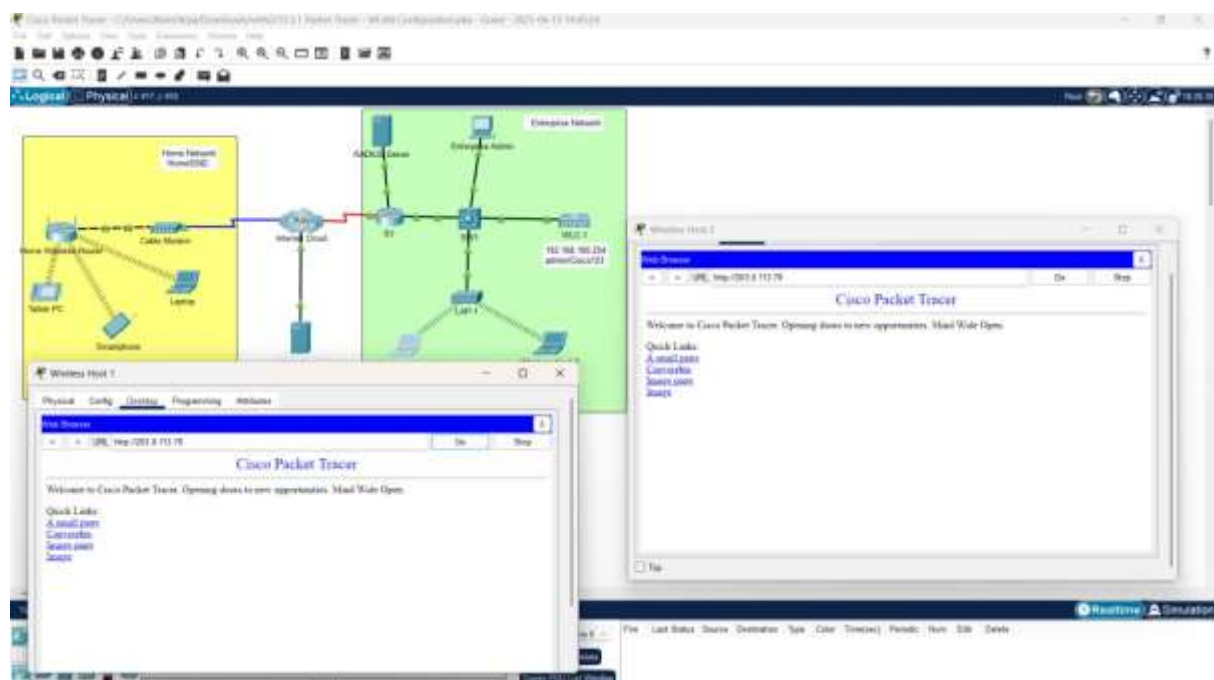


STEP 6: Testing connectivity

I tested connectivity between the wireless hosts and the web server by ping and URL Pinging was successful as shown:



The two hosts: Wireless Host 1 and Wireless Host 2 were also able to access the server through their browsers as shown below:



Conclusion

Through this assignment, I gained valuable hands-on experience in configuring and securing wireless networks in both home and enterprise environments. I learned how to modify DHCP settings, enable SSID broadcasting, and apply WPA2 security to protect wireless communications. I also became familiar with enterprise-grade configurations such as VLAN interfaces, WLC management, DHCP scopes, and RADIUS authentication. Testing network

connectivity and observing successful client associations helped reinforce my grasp of these concepts. Overall, the assignment deepened my practical knowledge of wireless LAN configuration and the importance of implementing robust security measures in network design